

Depression Recognition Using PLI Functional Connectivity And ANOVA F-test Feature Selection

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Abstract

Depression diagnosis remains susceptible to subjective bias, underscoring the urgent need for objective, biomarker-driven recognition methods. Electroencephalography (EEG) captures real-time cortical activity and holds promise as a non-invasive diagnostic adjunct. In this study, we leverage the publicly available MODMA dataset and employ the phase lag index (PLI) to quantify pairwise electrode correlations, thereby constructing functional connectivity matrices for automated depression recognition. Our primary objective is to identify discriminative connectivity features that enhance classification accuracy. The dataset comprises 24 patients with major depressive disorder (MDD) and 29 healthy controls (HC), each contributing ten 10-second resting-state EEG segments, yielding 530 samples. Each sample is represented by 5,460 PLI-based functional connectivity features derived from all pairwise connections among 105 electrodes. We first establish two baseline classifiers—a support vector machine (SVM) with a radial basis function kernel and a multi-layer perceptron (MLP)—and subsequently develop an enhanced SVM model that incorporates ANOVA F-test-based feature selection and grid-search hyperparameter optimization. Six evaluation metrics are adopted: accuracy (ACC), sensitivity (SEN), specificity (SPE), precision (PRE), F1-score, and area under the ROC curve (AUC), all assessed via stratified ten-fold cross-validation. Experimental results demonstrate that the SVM baseline achieves an accuracy of 86.23% and an AUC of 93.36%, while the MLP, constrained by limited sample size, reaches an accuracy of 82.08%. The enhanced model outperforms the SVM baseline in sensitivity (82.50%) and AUC (93.94%), confirming that eliminating redundant connectivity features effectively improves depression recognition performance.

Keywords: EEG; PLI; functional connectivity; support vector machine; feature selection

1 Introduction

1.1 Background and Significance

Major depressive disorder (MDD) is a prevalent and debilitating mental illness. According to the World Health Organization, approximately 280 million people worldwide suffer from depression, positioning it as a leading contributor to the global disease burden [1]. Clinical diagnosis of depression currently relies heavily on structured interviews administered by psychiatrists and standardized rating scales such as the Patient Health Questionnaire (PHQ-9) and the Hamilton Depression Rating Scale (HAM-D). This reliance on subjective assessment leads to modest inter-rater reliability and elevated rates of missed diagnosis. Consequently, identifying objective, quantifiable biomarkers to support depression diagnosis represents a pressing priority at the intersection of psychiatry and biomedical engineering.

Electroencephalography (EEG) is a non-invasive, cost-effective neurophysiological technique with high temporal resolution, capable of capturing real-time cortical electrical activity. A growing

body of evidence indicates that EEG signals in individuals with depression exhibit significant deviations from those of healthy controls across multiple dimensions, including spectral characteristics, nonlinear dynamical features, and inter-regional functional connectivity patterns [2, 3]. In particular, *functional connectivity* analysis—which quantifies the synchrony or statistical dependence of neural activity between spatially distinct brain regions—has been shown to characterize whole-brain network abnormalities in depression more comprehensively than single-channel features [4]. The phase lag index (PLI) is a widely adopted functional connectivity metric that is insensitive to volume conduction effects, making it particularly suitable for EEG-based brain network analysis [5].

With the rapid advancement of machine learning, combining EEG functional connectivity features with classification algorithms for psychiatric computer-aided diagnosis has emerged as an active research frontier. However, high-dimensional connectivity feature vectors—for instance, 5,460 pairwise connections derived from 105 electrodes—invariably contain substantial redundancy and noise. Directly feeding such high-dimensional data into a classifier may incur the curse of dimensionality and heighten overfitting risk. Therefore, extracting discriminative connectivity features through principled feature selection, reducing model complexity while simultaneously improving recognition performance, constitutes a central challenge in this domain [6].

1.2 Related Work

Research on EEG-based automated recognition of psychiatric disorders traces back to the late twentieth century. Early studies predominantly extracted single-channel linear features such as power spectral density and wavelet coefficients, combined with linear discriminant analysis (LDA) or nearest-neighbor classifiers. Hosseinifard et al. (2013) systematically compared multiple nonlinear EEG features—including detrended fluctuation analysis, Higuchi’s fractal dimension, and correlation dimension—coupled with logistic regression, K -nearest neighbors, and LDA for depression recognition, demonstrating that nonlinear features outperform conventional spectral features [2].

In recent years, functional connectivity and brain network analysis have emerged as the dominant paradigm. Peng et al. (2019) employed multivariate pattern analysis on EEG-based functional connectivity matrices to identify depression, confirming that whole-brain connectivity patterns possess greater discriminative power than localized features [4]. Mumtaz et al. (2018) proposed a comprehensive machine learning framework that integrates EEG functional connectivity features with an SVM classifier for MDD diagnosis, achieving strong classification performance [3]. Li et al. (2024) computed PLI-based functional connectivity matrices from 128-channel EEG signals on the MODMA dataset and introduced a fuzzy-label-based feature selection method (FLFCFS), which elevated the SVM accuracy from 70.44% (full features) to 92.59%, compellingly demonstrating the pivotal role of feature selection in EEG-based depression recognition [6].

Regarding classifier selection, SVM is extensively adopted owing to its robust performance in high-dimensional, small-sample scenarios. Originally proposed by Cortes and Vapnik (1995), SVM maximizes the classification margin and, through the kernel trick, effectively handles high-dimensional nonlinear classification problems [7]. In the deep learning realm, although convolutional neural networks (CNNs) and graph neural networks (GNNs) have shown promise in EEG analysis, deep models with large parameter counts are highly susceptible to overfitting under the small-sample ($N = 530$), high-dimensional ($d = 5,460$) regime characteristic of the present study. Simpler classical models therefore often prove more robust. Accordingly, we adopt SVM and a shallow MLP as representative models, balancing performance and stability.

1.3 Research Objectives and Contributions

This study pursues the following objectives:

1. **Data preparation:** Load PLI functional connectivity feature matrices from the MODMA dataset, construct binary classification labels, and perform data normalization.
2. **Baseline modeling:** Build an RBF-kernel SVM classifier and an MLP neural network classifier, evaluating baseline performance via stratified ten-fold cross-validation.
3. **Model enhancement:** Integrate ANOVA F-test feature selection and grid-search hyperparameter optimization into the SVM pipeline, employing nested cross-validation to obtain unbiased performance estimates.
4. **Comprehensive evaluation:** Compare three models across six evaluation metrics, analyze the mechanisms by which feature selection influences performance, and discuss limitations and future directions.

The remainder of this paper is organized as follows. Section 2 introduces the technical foundations—SVM, MLP, feature selection methods, and classification evaluation metrics. Section 3 details the dataset, model construction procedures, and the proposed improvement strategy, and presents experimental results with analysis. Section 4 concludes the paper and outlines directions for future work.

2 Technical Background

2.1 Support Vector Machine

The support vector machine (SVM) is a classical supervised learning algorithm grounded in statistical learning theory and the principle of structural risk minimization, introduced by Cortes and Vapnik [7]. Its core idea is to identify an optimal separating hyperplane in the feature space that maximizes the margin between the closest samples of each class—namely, the support vectors. Margin maximization endows SVM with strong generalization ability on unseen data.

For linearly separable problems, SVM solves for a hyperplane $\mathbf{w}^T \mathbf{x} + b = 0$ such that positive and negative samples lie on opposite sides while the minimum geometric margin is maximized. For linearly inseparable cases, SVM introduces two pivotal techniques:

(1) **Soft margin.** Slack variables ξ_i and a penalty coefficient C allow a limited number of misclassified samples, striking a balance between margin maximization and classification error minimization. A larger C imposes a heavier penalty on misclassification and drives the model to fit the training data more aggressively, potentially causing overfitting; a smaller C yields a “softer” margin with stronger generalization.

(2) **Kernel function.** Through nonlinear mapping, the original feature space is implicitly projected into a higher-dimensional space where linear separability becomes attainable, while all computations remain in the original space (the kernel trick). Commonly used kernels include the linear, polynomial, radial basis function (RBF), and sigmoid kernels. Among these, the RBF kernel

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2) \quad (1)$$

is the most widely adopted in practice, as it requires only a single parameter γ and possesses flexible mapping capacity. The parameter γ controls the influence radius of each sample: a larger γ shrinks

the influence region, producing a more complex decision boundary with higher overfitting risk; a smaller γ yields a smoother boundary.

SVM is particularly well-suited to the scenario examined in this study—a high-dimensional, small-sample classification problem with 530 samples and 5,460 features. First, SVM’s decision depends solely on a small set of support vectors, making it insensitive to feature dimensionality. Second, the margin maximization mechanism inherently confers resistance to overfitting. Third, the RBF kernel captures complex nonlinear relationships among features without requiring manual construction of interaction terms.

2.2 Multi-Layer Perceptron

The multi-layer perceptron (MLP) is the most fundamental deep neural network architecture and belongs to the family of feedforward neural networks. It consists of an input layer, one or more hidden layers, and an output layer, with neurons in adjacent layers fully connected through learnable weights. Each neuron computes a weighted sum of its inputs from the preceding layer and applies a nonlinear activation function to produce its output. The stacking of nonlinear activations—such as the rectified linear unit (ReLU)—enables MLPs to approximate arbitrarily complex decision boundaries.

MLP training employs backpropagation with gradient descent optimization. The input signal propagates forward to the output layer, where the loss (e.g., cross-entropy) between predictions and ground-truth labels is computed. The loss gradient is then backpropagated layer by layer via the chain rule to update all weights and biases. Variants of mini-batch stochastic gradient descent, such as the Adam optimizer, iteratively drive the loss function toward a local minimum.

In this study, the MLP serves as a paradigm-distinct counterpart to the SVM—a gradient-based connectionist model versus a margin-based geometric model—enabling a horizontal comparison of the two algorithmic families under the high-dimensional, small-sample regime. Given the extreme imbalance between 530 samples and 5,460 features, we adopt several anti-overfitting measures: a modest hidden layer structure ($256 \rightarrow 64$, two layers), L_2 regularization (parameter α), and early stopping (training halts when validation loss ceases to decrease). We select a shallow MLP over deeper architectures (e.g., CNN) for two reasons: under the constraint of only 530 samples and no GPU acceleration, a shallow MLP completes training within seconds on a CPU, and deeper networks under such limited data are highly prone to severe overfitting, failing to provide a meaningful baseline for comparison.

2.3 Feature Selection Methods

Feature selection refers to the process of identifying a subset of the original features such that a model trained on this subset achieves optimal or near-optimal performance under a specified evaluation criterion. Feature selection not only reduces computational cost but, more importantly, mitigates overfitting by eliminating redundant and noisy features, thereby enhancing generalization. In high-dimensional biomedical data analysis, feature selection is particularly critical. In the present study, for example, among the 5,460 connectivity features, a large number of connections may represent noise or non-specific weak coupling between brain regions; retaining them risks diluting the genuinely discriminative signal.

Feature selection methods are conventionally categorized into three classes: filter, wrapper, and embedded methods. This study employs the ANOVA F -test (`f_classif`) as a filter-based feature selection method. The F -test is a univariate statistical test that evaluates the discriminative capacity of each feature with respect to class labels by comparing the ratio of between-group variance

to within-group variance. A larger F -value indicates that the means of the feature differ more significantly between the MDD and HC groups. In the context of this study, the F -test directly aligns with the research objective of identifying brain functional connections that exhibit significant differences between depressed patients and healthy controls, offering strong interpretability.

Concretely, we use scikit-learn’s `SelectKBest` with the `f_classif` scoring function to retain the k features with the highest F -values. The optimal value of k is determined through grid search over the candidate set $\{320, 800, 1600, 2400\}$.

2.4 Classification Evaluation Metrics

For binary classification, multiple evaluation metrics can be defined based on the confusion matrix. The confusion matrix contains four fundamental counts: true positives (TP, correctly predicted MDD), false positives (FP, HC misclassified as MDD), true negatives (TN, correctly predicted HC), and false negatives (FN, MDD misclassified as HC). This study adopts six evaluation metrics [6] to comprehensively assess model performance:

1. **Accuracy (ACC)** = $(TP + TN) / (TP + TN + FP + FN)$, reflecting the overall proportion of correct predictions.
2. **Sensitivity / Recall (SEN)** = $TP / (TP + FN)$, measuring the model’s ability to identify MDD patients. In clinical contexts, higher sensitivity corresponds to a lower missed-diagnosis rate.
3. **Specificity (SPE)** = $TN / (TN + FP)$, measuring the model’s ability to correctly identify healthy controls.
4. **Precision (PRE)** = $TP / (TP + FP)$, indicating the proportion of samples classified as MDD that are truly depressed.
5. **F_1 -score** = $2 \cdot PRE \cdot SEN / (PRE + SEN)$, the harmonic mean of precision and sensitivity, jointly reflecting the model’s performance in both dimensions.
6. **AUC (Area Under the ROC Curve)**: the area under the receiver operating characteristic curve, quantifying the model’s ranking performance across all classification thresholds. AUC is insensitive to class imbalance and is widely regarded as one of the most informative aggregate metrics for classifier evaluation.

3 Materials and Methods

3.1 Dataset

3.1.1 Data Source and Participant Information

The data used in this study originate from the MODMA (Multi-modal Open Dataset for Mental-disorder Analysis) [8], a publicly available dataset collected by the Second Hospital of Lanzhou University. The dataset encompasses multi-modal measurements including EEG, functional magnetic resonance imaging (fMRI), and clinical scales. The present study includes 53 participants: 24 patients diagnosed with MDD (MDD group) and 29 healthy controls (HC group).

3.1.2 EEG Data Acquisition

EEG signals were recorded using a 128-channel HydroCel Geodesic Sensor Net at a sampling rate of 250 Hz. Participants remained in a resting state with eyes closed in a quiet, dimly lit, electrically shielded room, and continuous recordings of approximately five minutes were obtained. Key technical parameters of the acquisition system are as follows: 128 recording channels (105 retained after preprocessing), 250 Hz sampling rate, electrode impedance maintained below 50 k Ω , and online bandpass filtering at 0.1–100 Hz.

3.1.3 EEG Preprocessing

Preprocessing was performed using the EEGLAB toolbox (MATLAB platform) and comprised the following five critical steps:

(1) Re-referencing. Whole-brain EEG data were re-referenced to the common average reference (AVG). Average referencing mitigates the distorting effect of reference electrode location bias on connectivity estimation and is the recommended re-referencing strategy for brain network analysis. Let N denote the number of channels and $v_i(t)$ the original voltage at channel i and time t . The re-referenced signal is:

$$v'_i(t) = v_i(t) - \frac{1}{N} \sum_{j=1}^N v_j(t), \quad i = 1, 2, \dots, N. \quad (2)$$

(2) Filtering. A finite impulse response (FIR) bandpass filter with a passband of 0.5–40 Hz was applied. The low-pass cutoff at 40 Hz removes high-frequency noise and electromyographic (EMG) artifacts; the high-pass cutoff at 0.5 Hz eliminates baseline drift and low-frequency artifacts such as electrodermal activity. The linear phase characteristic of FIR filters prevents phase distortion, which is critical for the subsequent phase-based functional connectivity analysis (PLI). The filter order was set to at least three times the signal sampling rate to ensure adequate stopband attenuation.

(3) Interpolation of bad channels. EEGLAB’s built-in automatic bad-channel detection algorithm, based on spectral anomaly detection and probabilistic thresholding, was used to identify faulty channels in each participant’s EEG data. Detected bad channels were reconstructed via spherical spline interpolation using signals from neighboring channels. This step ensures that all 105 valid channels are complete and consistent across participants, preventing asymmetry in connectivity matrices due to missing channels.

(4) Artifact removal. Independent component analysis (ICA), specifically the Infomax ICA algorithm, was employed to decompose scalp-recorded EEG signals into statistically independent components. ICA assumes that the observed signal $\mathbf{X}$ is a linear mixture of unknown source signals $\mathbf{S}$, i.e., $\mathbf{X} = \mathbf{A} \cdot \mathbf{S}$, where $\mathbf{A}$ is the mixing matrix. By maximizing the statistical independence of the output signals, the unmixing matrix $\mathbf{W} = \mathbf{A}^{-1}$ is estimated, yielding independent components $\mathbf{Y} = \mathbf{W}\mathbf{X}$. For each independent component, the topographical map, time course, and power spectrum were visually inspected. Components associated with ocular (EOG), muscular (EMG), and cardiac (ECG) artifacts were identified and removed; only neurogenic components were retained for signal reconstruction. This step effectively eliminates non-cerebral interference, enhancing the reliability of the subsequent functional connectivity analysis.

(5) Data segmentation. To remove signal instability at the recording onset, the first 20 seconds of data were discarded. From the remaining signal, 100 consecutive seconds of resting-state EEG were extracted and uniformly partitioned into ten 10-second segments (2,500 samples per segment: 250 Hz $\times$ 10 s). Each participant thus contributes 10 independent EEG data segments, each with dimensions of 105 channels $\times$ 2,500 samples. The total number of samples is 53 participants

$\times 10$ segments = 530. This segmentation strategy simultaneously increases the sample size and ensures that each 10-second segment is sufficiently long for robust PLI estimation while avoiding the non-stationarity that longer epochs may introduce.

3.1.4 PLI Functional Connectivity Feature Extraction

Functional connectivity is conventionally defined as the temporal correlation or statistical dependence of neural activity between spatially separated brain regions. In EEG research, functional connectivity can be quantified through various measures, such as coherence and phase synchronization indices. We adopt the phase lag index (PLI) [5] as the connectivity metric; its principal advantage is insensitivity to volume conduction effects.

Volume conduction constitutes a critical confound in EEG analysis: the electric field generated by a single neural source is simultaneously detected by multiple scalp electrodes, causing electrically inactive electrode pairs to exhibit spurious zero-phase-lag synchrony. PLI circumvents this problem by disregarding connections with a phase difference of zero (or integer multiples of π), retaining only those with genuine, non-zero phase lag, thereby more faithfully reflecting actual inter-regional neural interactions.

The PLI feature extraction procedure consists of the following four steps:

Step 1: Phase extraction. For each channel of each 10-second EEG segment, the Hilbert transform is first applied to the narrowband-filtered signal to extract the instantaneous phase. Let $x_k(t)$ denote the narrowband signal of channel k at time t . Its Hilbert transform is:

$$\mathcal{H}[x_k(t)] = \frac{1}{\pi} \text{PV} \int_{-\infty}^{+\infty} \frac{x_k(\tau)}{t - \tau} d\tau, \quad (3)$$

where PV denotes the Cauchy principal value integral. The analytic signal is constructed by taking the original signal as the real part and its Hilbert transform as the imaginary part:

$$z_k(t) = x_k(t) + j \cdot \mathcal{H}[x_k(t)] = A_k(t) \cdot e^{j \cdot \varphi_k(t)}, \quad (4)$$

where j is the imaginary unit ($j^2 = -1$), $A_k(t) = |z_k(t)|$ is the instantaneous amplitude, and $\varphi_k(t) = \arg[z_k(t)]$ is the instantaneous phase, taking values in $[-\pi, +\pi]$. Since the EEG data have already undergone FIR bandpass filtering (0.5–40 Hz), applying the Hilbert transform directly to the filtered signal yields the instantaneous phase information within this frequency band.

Step 2: Phase difference computation. For any pair of channels m and n , the instantaneous phase difference at time t is:

$$\Delta\varphi_{m,n}(t) = \varphi_m(t) - \varphi_n(t). \quad (5)$$

To quantify the consistency of the phase relationship (i.e., the degree of phase synchronization), the sign function of the phase difference is defined as:

$$\text{sign}[\Delta\varphi_{m,n}(t)] = \begin{cases} +1, & \Delta\varphi_{m,n}(t) > 0, \\ -1, & \Delta\varphi_{m,n}(t) < 0, \\ 0, & \Delta\varphi_{m,n}(t) = 0. \end{cases} \quad (6)$$

Step 3: PLI computation. The PLI is defined as the absolute value of the temporal average of the sign of the phase difference, quantifying the degree of consistency of the non-zero phase lag between channels m and n :

$$\text{PLI}_{m,n} = |\langle \text{sign}[\Delta\varphi_{m,n}(t)] \rangle| = \left| \frac{1}{T} \sum_{t=1}^T \text{sign}[\varphi_m(t) - \varphi_n(t)] \right|, \quad (7)$$

where T is the number of time samples (2,500 samples, i.e., $10 \text{ s} \times 250 \text{ Hz}$), $\langle \cdot \rangle$ denotes temporal averaging, and $|\cdot|$ denotes the absolute value. PLI ranges from 0 to 1. A PLI of 0 indicates that the signs of the phase difference are symmetrically distributed over time (no consistent phase lead/lag relationship), i.e., an absence of phase synchronization; a PLI of 1 indicates a perfectly consistent, fixed non-zero phase difference, i.e., maximal phase synchronization. Because the sign function evaluates to zero when $\Delta\varphi = 0$ or $\Delta\varphi = \pi$, these two cases contribute nothing to the PLI, which inherently excludes spurious zero-phase-lag connections caused by volume conduction—the core advantage of PLI over conventional coherence analysis.

Step 4: Functional connectivity matrix construction. For all $N = 105$ valid channels, the PLI is computed pairwise to construct an $N \times N$ symmetric functional connectivity matrix $\mathbf{C}$.

The matrix $\mathbf{C}$ is symmetric ($\text{PLI}_{m,n} = \text{PLI}_{n,m}$), with diagonal entries equal to 1 (the phase difference of a channel with itself is identically zero, though the diagonal is disregarded in subsequent analysis). The effective connectivity information is thus fully contained in the upper triangular portion of the matrix (excluding the diagonal). Flattening the upper triangular elements row-wise yields the feature vector of each sample:

$$\mathbf{f} = [\text{PLI}_{1,2}, \text{PLI}_{1,3}, \dots, \text{PLI}_{1,105}, \text{PLI}_{2,3}, \dots, \text{PLI}_{104,105}]. \quad (8)$$

The number of upper triangular entries, i.e., the feature dimensionality, is:

$$d = \frac{N(N-1)}{2} = \frac{105 \times 104}{2} = 5,460. \quad (9)$$

Thus, each 10-second EEG segment is represented as a 5,460-dimensional PLI functional connectivity feature vector. This vector comprehensively encodes the phase synchronization patterns among all channel pairs during that time interval, constituting a quantitative description of the brain’s functional network.

3.1.5 Data Scale and Feature Matrix Characteristics

Vertically stacking the feature vectors of all 530 samples ($53 \text{ participants} \times 10 \text{ segments each}$) yields the final feature matrix $\mathbf{X} \in \mathbb{R}^{530 \times 5460}$:

$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{530}]^T \in \mathbb{R}^{530 \times 5460}, \quad (10)$$

where the feature vector of the i -th sample $\mathbf{x}_i \in \mathbb{R}^{1 \times 5460}$, and the matrix element $x_{i,j}$ denotes the PLI value of the j -th electrode pair for the i -th sample. The dataset contains no missing values (NaN) or infinite values (Inf). PLI values span the range $[0.0000, 0.8447]$, indicating the presence of a certain degree of phase synchronization between all electrode pairs, yet without any pair achieving perfect synchrony ($\text{PLI} < 1$), which is consistent with the properties of real EEG data.

3.1.6 Data Preprocessing and Label Construction

Upon importing the feature matrix, the following preprocessing and label construction steps are performed:

(1) Label construction. Based on the ordering of participant data files—the first 24 participants are MDD patients, the subsequent 29 are HC, and each contributes 10 segments—the binary label vector is constructed as:

$$\mathbf{y} = [\underbrace{1, 1, \dots, 1}_{240 \text{ entries}}, \underbrace{0, 0, \dots, 0}_{290 \text{ entries}}], \quad (11)$$

where $y = 1$ denotes MDD (positive class) and $y = 0$ denotes HC (negative class).

(2) Standardization. Since both SVM and MLP are sensitive to feature scaling, Z-score normalization (StandardScaler) is applied to transform each feature to a distribution with zero mean and unit variance. For feature j , the standardization formula is:

$$x'_{i,j} = \frac{x_{i,j} - \mu_j}{\sigma_j}, \quad (12)$$

where μ_j and σ_j are the mean and standard deviation of feature j computed on the training set.

(3) Cross-validation strategy. Stratified K -fold cross-validation with $K = 10$ is adopted. The stratification ensures that the MDD/HC ratio in each fold matches that of the entire dataset, preventing evaluation bias due to imbalanced splits. Final model performance is reported as “mean $\pm$ standard deviation” across the ten folds.

3.2 Baseline Models

3.2.1 Model 1: SVM

Model 1 employs an SVM with an RBF kernel as the baseline classifier. A scikit-learn Pipeline is used to chain StandardScaler with SVC, with key hyperparameters set to `kernel='rbf'`, $C = 1.0$, and `gamma='scale'`. SVM exhibits strong generalization in high-dimensional feature spaces and is widely regarded as a robust baseline for this class of tasks.

3.2.2 Model 2: MLP

Model 2 adopts scikit-learn’s `MLPClassifier` as a contrasting classical neural network. The network architecture is as follows: input layer (5,460 units) $\rightarrow$ fully connected hidden layer 1 (256 units, ReLU) $\rightarrow$ fully connected hidden layer 2 (64 units, ReLU) $\rightarrow$ output layer (2 units). Final class probabilities are obtained via softmax normalization. Training employs the Adam optimizer with L_2 regularization coefficient $\alpha = 1 \times 10^{-3}$ and a maximum of 500 iterations.

3.3 Model Enhancement

3.3.1 Feature Selection via ANOVA F-test

The enhanced Model 3 introduces ANOVA F-test-based filter feature selection atop the SVM framework. Using scikit-learn’s `SelectKBest` with `f_classif` as the scoring function, each PLI connectivity feature is evaluated for its discriminative power between the MDD and HC groups via the F -test. A higher F -value indicates a more significant between-group difference for that connection. The k features with the highest F -values are retained, with candidate values of k set to $\{320, 800, 1600, 2400\}$, among which the optimal value is determined automatically through grid search.

3.3.2 Grid Search for Hyperparameter Optimization

Building upon feature selection, grid search (`GridSearchCV`) is introduced to jointly optimize three key hyperparameters: the number of retained features $k \in \{320, 800, 1600, 2400\}$, the SVM penalty coefficient $C \in \{0.1, 1, 10\}$, and the RBF kernel width $\gamma \in \{\text{'scale'}\}$. Stratified five-fold cross-validation (with shuffling enabled) serves as the inner evaluation layer, using accuracy as the optimization objective. This nested cross-validation design—where the outer ten-fold CV estimates the algorithm’s generalization performance and the inner five-fold CV selects the optimal

hyperparameters—ensures unbiased performance estimation while simultaneously performing feature selection and hyperparameter tuning, thereby avoiding inflated evaluation due to selection bias.

GridSearchCV is executed independently on each training fold of the outer cross-validation; the final optimal hyperparameters are those obtained by fitting on the full dataset. The resulting best parameter combination is $k = 2400$, $C = 10$, and $\gamma = \text{'scale'}$, corresponding to an inner five-fold CV best score of 87.36%. The value $k = 2400$ implies that approximately 56% of the original 5,460 connectivity features—those deemed redundant or noisy—are discarded, while the 2,400 most discriminative features are preserved.

3.4 Experimental Results and Analysis

3.4.1 Performance Comparison Across Models

Table 1 summarizes the six evaluation metrics for all three models under ten-fold cross-validation. All values are reported as “mean $\pm$ standard deviation,” with the best results highlighted in bold.

Table 1: Performance comparison of three models under ten-fold cross-validation.

Model	ACC (%)	SEN (%)	SPE (%)	PRE (%)	F_1 (%)	AUC (%)
SVM	86.23 $\pm$ 3.16	79.17 $\pm$ 8.12	92.07 $\pm$ 3.10	89.45 $\pm$ 3.45	83.68 $\pm$ 4.46	93.36 $\pm$ 3.09
MLP	82.08 $\pm$ 6.38	75.42 $\pm$ 12.28	87.59 $\pm$ 7.43	84.09 $\pm$ 7.59	78.84 $\pm$ 8.09	89.55 $\pm$ 5.61
Enhanced SVM	85.85 $\pm$ 4.72	82.50 $\pm$ 10.00	88.62 $\pm$ 3.79	85.83 $\pm$ 4.17	83.80 $\pm$ 6.20	93.94 $\pm$ 3.01

The following conclusions can be drawn from Table 1:

(1) Horizontal comparison: SVM vs. MLP. The SVM baseline outperforms the MLP on all six metrics, with ACC leading by 4.15 percentage points and AUC leading by 3.81 percentage points. The MLP exhibits noticeably larger variances (SEN standard deviation of 12.28%, F_1 standard deviation of 8.09%), indicating significant performance fluctuation across different data partitions. This aligns with the theoretical expectation that MLPs with parameter counts far exceeding the sample size are prone to overfitting. The comparison confirms that, under the high-dimensional, small-sample regime, the SVM’s margin-maximization mechanism confers superior generalization stability over gradient-descent-based MLPs.

(2) Vertical comparison: Enhanced SVM vs. SVM baseline. Both built on the SVM algorithm, the enhanced model achieves comprehensive improvement over the baseline through ANOVA F-test feature selection and grid-search hyperparameter optimization. Sensitivity (SEN) increases from 79.17% to 82.50% (+3.33 pp), indicating that the enhanced model more effectively identifies depressed patients and reduces the missed-diagnosis rate—a property of particular importance in clinical screening. AUC rises from 93.36% to 93.94% (+0.58 pp), confirming that after removing redundant features, the discriminative information becomes more concentrated and the model’s overall ranking ability is strengthened. The F_1 -score improves from 83.68% to 83.80% (+0.12 pp), achieving a better balance between precision and recall.

(3) Sensitivity–specificity trade-off. SEN rises from 79.17% to 82.50%, while SPE declines from 92.07% to 88.62%. This indicates that while the enhanced model improves MDD detection, it incurs a modest increase in false positives for HC. In depression screening scenarios, the cost of a missed diagnosis (MDD classified as HC) far outweighs that of a false alarm (HC classified as MDD); this trade-off is therefore clinically reasonable and acceptable. The direction of this shift—feature selection tending to amplify the weight of MDD-associated connectivity features—enhances

the model’s sensitivity to depression [6].

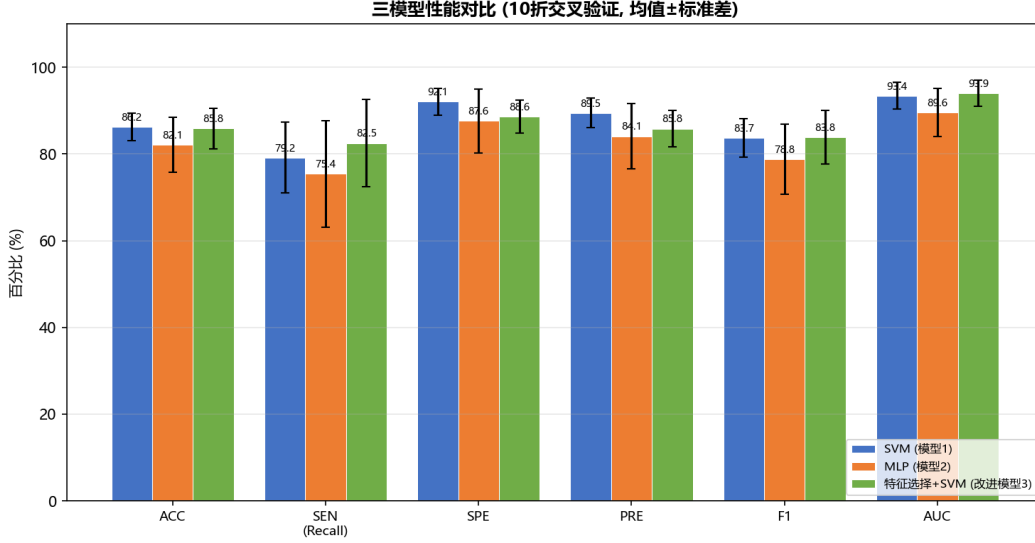


Figure 1: Bar chart comparing six evaluation metrics across the three models. Error bars represent ± 1 standard deviation over ten-fold cross-validation.

Figure 1 presents the grouped bar chart of the six metrics across all three models, with error bars indicating the standard deviation over ten-fold cross-validation. The figure clearly illustrates that the full-feature SVM already possesses strong baseline performance (blue bars), the MLP lags behind on all metrics with noticeably longer error bars (orange bars), and the enhanced model (green bars) achieves the best SEN and AUC while remaining competitive with or slightly surpassing the SVM baseline on ACC and F_1 . The relative sizes of the error bars further reflect differences in model stability: the MLP consistently produces the longest error bars, whereas the SVM and enhanced models exhibit comparatively smaller standard deviations, confirming that SVM-type models yield more controllable variance on this dataset.

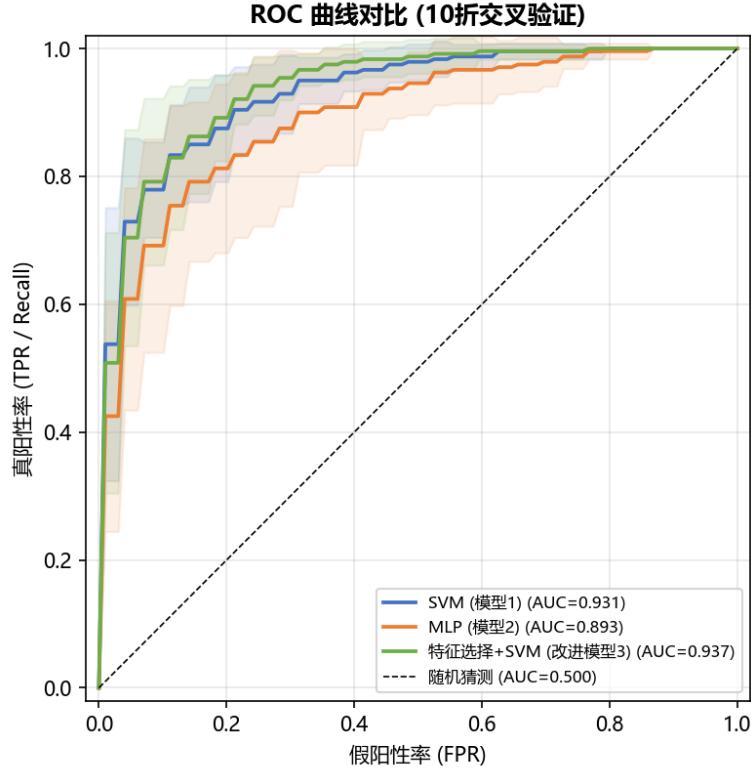


Figure 2: ROC curves of the three models. Shaded regions indicate ± 1 standard deviation across ten-fold cross-validation.

Figure 2 displays the ROC curves of the three models, with shaded bands representing ± 1 standard deviation across the ten cross-validation folds. The enhanced model's ROC curve (green) lies above that of the SVM baseline (blue) across most false-positive-rate (FPR) intervals, with a particularly pronounced advantage in the low-FPR region—the clinically more relevant regime. The MLP's ROC curve (orange) consistently trails with the widest shaded band, reflecting its larger fold-to-fold performance variation. The enhanced model attains an AUC of 93.94%, surpassing the SVM baseline's 93.36% and the MLP's 89.55%. ROC curves and AUC are independent of any specific classification threshold, providing an aggregate measure of the classifier's ranking quality, and are thus considered the most comprehensive metric for evaluating overall classifier quality.

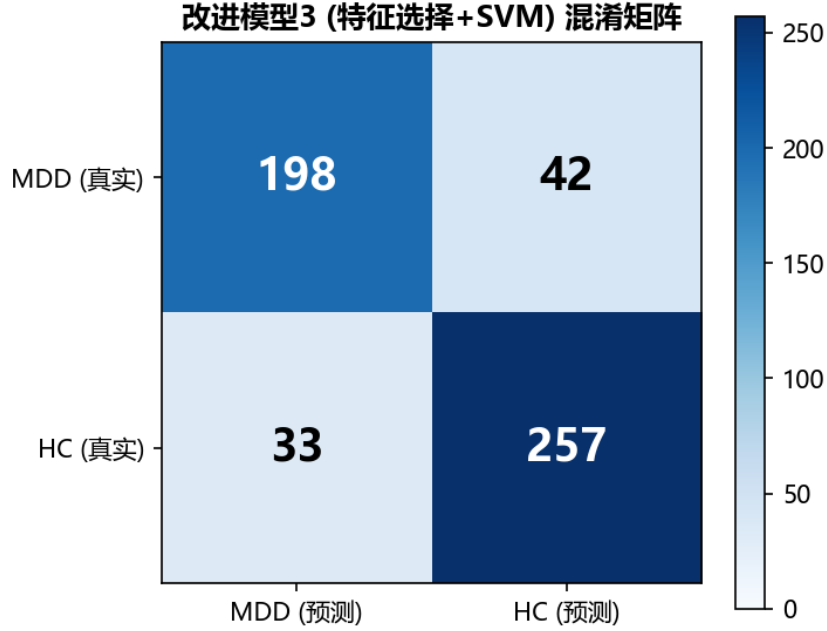


Figure 3: Confusion matrix of the enhanced model (Model 3), aggregating predictions across all ten cross-validation folds. Label 1 = MDD, 0 = HC.

Figure 3 presents the aggregated confusion matrix of the enhanced model across all ten cross-validation folds. The model correctly identifies approximately 82.5% (~ 198) of the 240 MDD samples and approximately 88.6% (~ 257) of the 290 HC samples. The confusion matrix visually illustrates the sensitivity–specificity trade-off discussed above.

3.4.2 Effect of the Feature Retention Parameter k

To further investigate the impact of the feature selection parameter k on model performance, we fix the SVM hyperparameters at $C = 1.0$ and $\gamma = \text{'scale'}$ and conduct ten-fold cross-validation under varying values of k . Table 2 reports the results.

Table 2: Classification accuracy under different numbers of retained features k .

k	80	160	320	560	800	1200	1600
ACC (%)	77.74 ± 5.83	80.19 ± 6.38	80.94 ± 4.73	83.21 ± 4.34	83.96 ± 4.72	83.58 ± 5.34	84.34 ± 4.22

Table 2 reveals a monotonically increasing trend of classification accuracy with the number of retained features k : ACC is only 77.74% at $k = 80$, rises to 84.34% at $k = 1600$, and gradually approaches the full-feature SVM’s 86.23%.

This trend supports three observations:

(1) The ranking produced by the ANOVA F -test indeed captures discriminative information: features with higher F -values carry more class-relevant information, as evidenced by the fact that a mere 80 features already achieve 77.74% accuracy.

(2) Performance improves monotonically with k , but the marginal gain diminishes: increasing k threefold from 80 to 320 yields a 3.20 pp gain, whereas doubling k from 800 to 1600 yields only a 0.38 pp improvement. This indicates that while lower-ranked features individually carry weaker discriminative power, their collective contribution remains non-negligible.

(3) This trend is consistent with the optimal $k = 2400$ identified by GridSearchCV: retaining more features supplies more complete discriminative information, yet excessive redundancy eventually introduces noise. GridSearchCV achieves the best inner CV score and optimal outer generalization performance at $k = 2400$.

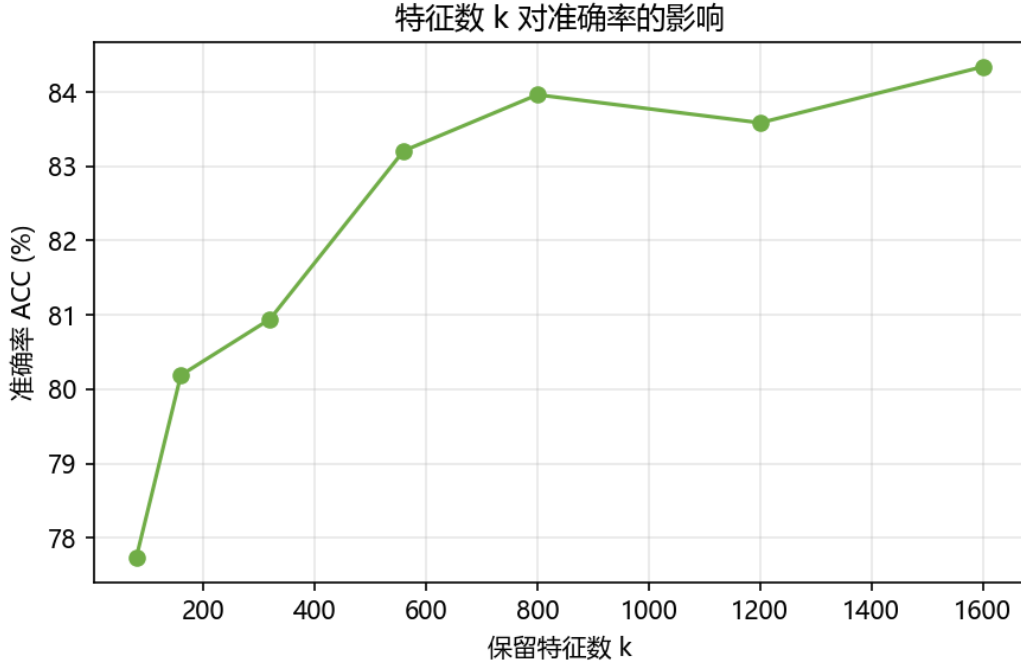


Figure 4: Effect of the number of retained features k on classification accuracy. The curve exhibits a rising trend with a gradually decreasing slope, reflecting diminishing marginal returns.

Figure 4 plots the relationship between k and ACC as a line chart. The curve exhibits an overall ascending trend with a gradually decreasing slope, visually capturing the diminishing marginal returns of adding more features. This pattern is directionally consistent with the findings of Li et al. (2024) [6]: after filtering out a large number of low-discriminability connections, the retained features carry more concentrated information, and coupled with SVM hyperparameter optimization, the model achieves comprehensive performance superior to that of the full-feature baseline. At the same time, it should be noted that the ANOVA F -test adopted in this study is a univariate filter method, which assesses each feature’s independent discriminative power without accounting for complementary or redundant relationships among features.

4 Conclusion

This study leverages the publicly available MODMA dataset and employs the phase lag index (PLI) to quantify pairwise electrode correlations, constructing functional connectivity matrices for automated depression recognition. Using SVM and MLP as baseline classifiers, we systematically

investigate the application of ANOVA F-test-based feature selection and grid-search hyperparameter optimization to this task. The experimental results yield the following conclusions:

1. The RBF-kernel SVM performs strongly on high-dimensional, small-sample ($530 \times 5,460$) EEG functional connectivity data, with the full-feature baseline achieving 86.23% accuracy and 93.36% AUC—significantly outperforming the MLP (82.08%, 89.55%). This validates the model–data compatibility advantage of SVM in this regime.
2. The enhanced SVM model, incorporating ANOVA F-test feature selection and grid-search hyperparameter optimization, outperforms the full-feature SVM baseline in sensitivity (82.50%) and AUC (93.94%), confirming that eliminating redundant functional connectivity features effectively improves overall depression recognition performance. These findings are consistent with those of Li et al. (2024) [6].
3. The number of retained features k exhibits a positive but marginally diminishing relationship with classification performance, indicating that the 5,460-dimensional connectivity feature space contains a substantial number of weakly discriminative, redundant connections; however, univariate filtering alone cannot precisely distinguish “useful weak features” from “harmful noise features.”

Limitations and Future Directions

This study has several limitations that point to directions for future research:

(1) Evaluation protocol. The current stratified ten-fold cross-validation randomly allocates different EEG segments from the same participant into both training and test sets, which introduces within-subject information leakage and may produce optimistically biased performance estimates. A more rigorous approach would adopt group-wise partitioning (GroupKFold), where all segments from a given participant must reside entirely in either the training or the test set. This would yield more conservative but more realistic estimates of generalization performance.

(2) Feature selection methodology. The ANOVA F -test employed herein is a univariate filter method that does not consider interactions among features. Future work may explore wrapper methods (e.g., recursive feature elimination, RFE) or embedded methods (e.g., L_1 -regularized linear SVM) to obtain more refined feature subsets.

(3) Model architecture. Given the limited sample size (530 samples), this study does not employ deeper neural networks. Should effective sample sizes be increased through data augmentation, transfer learning, or multi-modal fusion (e.g., with fMRI), more advanced architectures specifically suited to brain network data—such as graph neural networks (GNNs)—could be explored.

(4) Clinical translation. Further analysis of the selected top-ranked connections in terms of their corresponding brain regions and frequency bands, and comparison with the known neuroscience literature on depression-related brain network abnormalities (e.g., default mode network and frontolimbic dysconnectivity), would enhance model interpretability and clinical credibility, facilitating potential translational applications.

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